

Problem Set 3: Solution

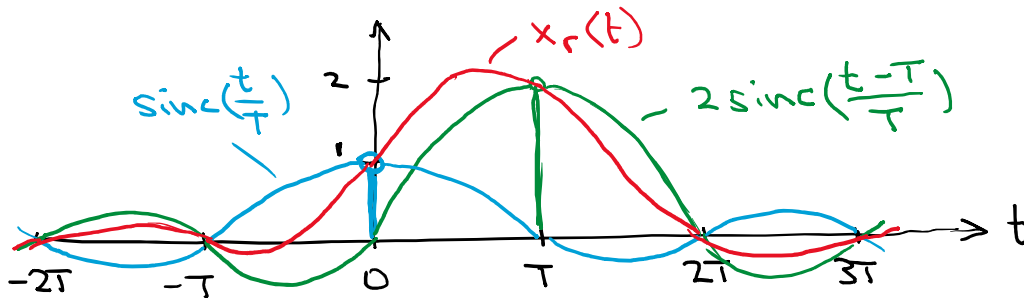
1. Ideal sinc reconstruction

Let the discrete-time sequence be $x[n] = \delta[n] + 2\delta[n - 1]$, obtained from a valid bandlimited sampling process with sample period T .

- Write the ideal continuous-time reconstructed signal $x_r(t)$ using normalized $\text{sinc}(u) = \sin(\pi u)/(\pi u)$.
- Verify $x_r(0) = 1$ and $x_r(T) = 2$.
- Compute $x_r(T/2)$.

$$x[n] = \delta[n] + 2\delta[n-1] \quad \text{ideally bandlimited } x(t)$$

$$a) x_r(t) = \text{sinc}\left(\frac{t}{T}\right) + 2 \text{sinc}\left(\frac{t-T}{T}\right)$$



$$b) x_r(0) = \text{sinc}(0) + 2 \text{sinc}(-1) = 1 \quad \checkmark$$

$$x_r(T) = \text{sinc}(1) + 2 \text{sinc}(0) = 2 \quad \checkmark$$

$$c) x_r\left(\frac{T}{2}\right) = \text{sinc}\left(\frac{1}{2}\right) + 2 \text{sinc}\left(-\frac{1}{2}\right) = \frac{6}{\pi} = 1.910$$

$$\text{sinc}\left(\frac{1}{2}\right) = \text{sinc}\left(-\frac{1}{2}\right) = \frac{\sin(\pi/2)}{\pi/2} = \frac{2}{\pi}$$

2. Zero-order hold frequency response

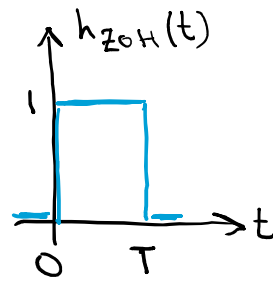
A DAC uses a zero-order hold $h_{ZOH}(t) = u(t) - u(t - T)$.

- Derive $H_{ZOH}(j\Omega)$ by direct integration.
- Separate the result into a linear-phase term and a sinc-like magnitude term.
- Compute the normalized magnitude $|H_{ZOH}|/T$ in dB at $\omega = \Omega T = 0.5\pi$ and at $\omega = \pi$.

a)

$$H_{ZOH}(j\Omega) = \int_0^T e^{-j\Omega t} dt$$

$$= \frac{e^{-j\Omega t}}{-j\Omega} \Big|_0^T = \frac{1 - e^{-j\Omega T}}{j\Omega}$$



$$b) H_{ZOH}(j\Omega) = \frac{e^{j\Omega T/2} - e^{-j\Omega T/2}}{2j\Omega/2} e^{-j\Omega T/2}$$

$$= T \frac{\sin(\frac{\Omega T}{2})}{T\Omega/2} e^{-j\Omega T/2} = T \text{sinc}(\frac{\Omega T}{2\pi}) e^{-j\Omega T/2}$$

sinc-like linear
magnitude phase term
term

$$c) \frac{1}{T} |H_{ZOH}(j\Omega)| \Big|_{\Omega T = \pi/2} = \text{sinc}(\frac{1}{4}) = \frac{4}{\sqrt{2}\pi} = 0.9003$$

$\omega = \pi/2$

$$\frac{1}{T} |H_{ZOH}(j\Omega)| \Big|_{\Omega T = \pi} = \text{sinc}(\frac{1}{2}) = \frac{2}{\pi} = 0.6366$$

$\omega = \pi$

3. Uniform quantization and SQNR

A uniform B-bit quantizer spans approximately -1 V to $+1$ V. Assume the high-resolution uniform-error model when requested.

- For $B = 8$, find the quantization step Δ .
- Find the modeled error variance $\sigma_e^2 = \Delta^2/12$ and RMS quantization error.
- Estimate the SQNR of a full-scale 8-bit sine.
- Estimate the full-scale sine SQNR for 16 bits.
- Explain why "16-bit audio is about 96 dB" and "16-bit full-scale sine SQNR is about 98 dB" are not necessarily contradictory

$$a) B=8, \Delta = \frac{2A}{2^B} = \frac{2}{256} = \frac{1}{128} = 0.0078125 \text{ V}$$

$$b) \sigma_e^2 = \frac{\Delta^2}{12} = \frac{1}{12 \times 2^{14}} = \frac{1}{3 \times 2^{16}} = 5.086 \times 10^{-6} \text{ V}^2$$

$$\text{RMS quantization error} = \sqrt{\sigma_e^2} = 2.255 \times 10^{-3} \text{ V}$$

$\Rightarrow$ step size $\Delta = 7.81 \text{ mV}$, RMS quant error 2.255 mV

$$c) \text{SQNR} = \frac{P_x}{P_e} = \frac{A^2/2 \times 12}{(2A/2^B)^2} = 98304 \hat{=} 49.92 \text{ dB}$$

$$d) B=16, \text{SQNR} = 6.02 \times 16 + 1.76 = 98.08 \text{ dB}$$

e) Two different models:

$$* \text{ Full scale sine } \text{SQNR} = 6.02 \cdot B + 1.76 \text{ dB}$$

$$P_x = A^2/2$$

$\approx 98 \text{ dB}$, see part d)

* Amplitude ratio over 16 bits (quantization error is divided by 2, energy by 4 for each additional bit)

$$\text{SQNR} = 16 \times 6.02 = 96.32 \text{ dB}$$

4. Finite sequence z-transform

Let $x[n] = 2\delta[n+1] - \delta[n] + 3\delta[n-2]$. Find $X(z)$, state the ROC, and determine whether the DTFT exists.

$$X(z) = 2z^{-1} + 3z^{-2}$$

ROC $0 < |z| < \infty$ (Both positive and negative powers of z appear and there is only a finite number of terms)

The unit circle ($|z|=1$) is included, so the DTFT exists.

5. Right-sided exponential and convergence

For $x[n] = (1.2)^n u[n]$, derive $X(z)$ from the geometric series, state the ROC, identify the pole, and determine whether its DTFT exists.

$$X(z) = \sum_n x[n] z^{-n} = \sum_{n=0}^{\infty} (1.2)^n z^{-n} = \frac{1}{1 - 1.2 z^{-1}}$$

$\uparrow$
 $|1.2 z^{-1}| < 1$
 $\Rightarrow \text{ROC } |z| > 1.2$

Pole at $z = 1.2$, zero at $z = 0$

$|z| = 1$ is not in ROC $\Rightarrow$ DTFT does not exist

6. Same algebra, different ROC

Find the inverse z-transform of $X(z) = 1/(1 - 0.75z^{-1})$ for

a) $|z| > 0.75$ and

b) $|z| < 0.75$.

Explain in one sentence why the two answers can have the same rational expression.

$$X(z) = \frac{1}{1 - 0.75z^{-1}}$$

a) $|z| > 0.75$ $x[n] = (0.75)^n u[n]$

$$X(z) = \sum_{n=0}^{\infty} (0.75)^n z^{-n} = \frac{1}{1 - 0.75z^{-1}}$$

↑
 $|0.75 \cdot z^{-1}| < 1 \Rightarrow \text{ROC } |z| > 0.75$

b) $|z| < 0.75$ $x[n] = -(0.75)^n u[-n-1]$

$$X(z) = - \sum_{n=-\infty}^{-1} (0.75)^n z^{-n} = - \sum_{m=1}^{\infty} (0.75)^{-m} z^m =$$

$-m \leftarrow n$
 $m \leftarrow -n$

$$= \frac{-1}{1 - (0.75)^{-1}z} + 1 = \frac{-(0.75)^{-1}z}{1 - (0.75)^{-1}z}$$

↑

$|z/0.75| < 1$

ROC $|z| < 0.75$

$$= \frac{-1}{0.75z^{-1} - 1} = \frac{1}{1 - 0.75z^{-1}}$$

Part a) is the expansion of $X(z)$ in terms of powers of z^{-1} (causal sequence) and part b) is the expansion of $X(z)$ in terms of powers of z (anticausal sequence).

7. Partial fractions with two ROCs (ECEN 5632 Problem)

Let $X(z) = 1/[(1 - 0.5z^{-1})(1 - 0.8z^{-1})]$. Use partial fractions and find $x[n]$ for

a) $|z| > 0.8$ and

b) $0.5 < |z| < 0.8$.

$$X(z) = \frac{1}{(1 - 0.5z^{-1})(1 - 0.8z^{-1})}$$
$$= \frac{A}{1 - 0.5z^{-1}} + \frac{B}{1 - 0.8z^{-1}}$$

$$A = \frac{1}{1 - 0.8z^{-1}} \Big|_{z^{-1}=2} = \frac{1}{1 - 1.6} = \frac{-5}{3}$$

$$B = \frac{1}{1 - 0.5z^{-1}} \Big|_{z^{-1}=\frac{5}{4}} = \frac{1}{1 - 5/8} = \frac{8}{3}$$

$$\Rightarrow X(z) = \frac{-5/3}{1 - 0.5z^{-1}} + \frac{8/3}{1 - 0.8z^{-1}}$$

a) $|z| > 0.8$

$$x[n] = \left[\frac{-5}{3} (0.5)^n + \frac{8}{3} (0.8)^n \right] u[n]$$

b) $0.5 < |z| < 0.8$

$$x[n] = \frac{-5}{3} (0.5)^n u[n] - \frac{8}{3} (0.8)^n u[-n-1]$$